

Technical Documentation

Wind Turbine Generator Systems

3.4-130 - 50/60 Hz



Calculated Power Curve and Thrust Coefficient

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1 Calculated Power Curve

Wind turbine type: 3.4-130
 Grid standard: 50 Hz and 60 Hz
 Rotor diameter: 130 m

The calculated power curve at an average air density of 1.225 kg/m³ is listed in Table 1. This power curve is consistent with the power curve definition in the IEC 61400-12-1 (2005) for hub height wind speed.

Turbulence intensity (TI) ranges are defined in section 3. The method for selecting the appropriate turbulence intensity is provided in section 6.

Wind Speed at Hub Height [m/s]	Electrical Power [kW] with			Cp,e Medium TI
	Medium TI Band	Low TI Band	High TI Band	
3.0	56	46	83	0.25
3.5	128	117	158	0.37
4.0	215	203	249	0.41
4.5	321	307	358	0.43
5.0	449	433	493	0.44
5.5	606	587	657	0.45
6.0	791	770	851	0.45
6.5	1010	985	1074	0.45
7.0	1263	1235	1334	0.45
7.5	1547	1521	1609	0.45
8.0	1863	1840	1915	0.45
8.5	2201	2189	2225	0.44
9.0	2531	2537	2504	0.43
9.5	2827	2854	2756	0.41
10.0	3058	3109	2950	0.38
10.5	3218	3275	3101	0.34
11.0	3335	3377	3239	0.31
11.5	3387	3414	3312	0.27
12.0	3414	3427	3359	0.24
12.5	3425	3430	3391	0.22
13.0	3429	3430	3407	0.19
13.5	3430	3430	3414	0.17
14.0	3430	3430	3420	0.15
14.5	3430	3430	3424	0.14
15.0	3430	3430	3427	0.13
15.5	3430	3430	3430	0.11
16.0	3430	3430	3430	0.10
16.5	3430	3430	3430	0.09
17.0	3430	3430	3430	0.09
17.5	3430	3430	3430	0.08
18.0	3430	3430	3430	0.07
18.5	3430	3430	3430	0.07
19.0	3430	3430	3430	0.06
19.5	3430	3430	3430	0.06

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Wind Speed at Hub Height [m/s]	Electrical Power [kW] with			Cp,e Medium TI
	Medium TI Band	Low TI Band	High TI Band	
20.0	3430	3430	3430	0.05
20.5	3430	3430	3430	0.05
21.0	3430	3430	3430	0.05
21.5	3430	3430	3430	0.04
22.0	3430	3430	3430	0.04
22.5	3430	3430	3430	0.04
23.0	3430	3430	3430	0.03
23.5	3430	3430	3430	0.03
24.0	3430	3430	3430	0.03
24.5	3430	3430	3430	0.03
25.0	3430	3430	3430	0.03

Table 1: Calculated power curve table as a function of hub height wind speed for standard air density

2 Power Curves for different Air Density and Turbulence Intensity

Following tables show the calculated power curves for different air density values and turbulence intensity classes. The excel file embedded with this document contains data for a larger range of air densities than in the tables below.

Wind Speed at Hub Height [m/s]	Air Density										
	$\rho = 1.02$	$\rho = 1.04$	$\rho = 1.06$	$\rho = 1.08$	$\rho = 1.1$	$\rho = 1.12$	$\rho = 1.14$	$\rho = 1.16$	$\rho = 1.18$	$\rho = 1.2$	$\rho = 1.225$
	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³
3.0	40	42	43	45	46	48	49	51	52	54	56
3.5	100	103	106	109	111	114	117	119	122	125	128
4.0	173	177	181	185	189	194	198	202	206	210	215
4.5	260	266	272	278	284	290	296	302	307	313	321
5.0	367	375	383	391	399	407	415	423	431	439	449
5.5	497	508	519	529	540	550	561	571	582	593	606
6.0	652	666	679	693	706	720	733	747	761	774	791
6.5	834	851	869	886	903	920	937	954	971	988	1010
7.0	1046	1067	1089	1110	1131	1152	1173	1194	1215	1236	1263
7.5	1289	1315	1340	1366	1392	1417	1442	1467	1492	1517	1547
8.0	1561	1592	1622	1653	1683	1712	1741	1769	1798	1827	1863
8.5	1863	1897	1931	1965	1999	2032	2065	2098	2131	2164	2201
9.0	2177	2214	2251	2288	2325	2359	2393	2428	2462	2496	2531
9.5	2488	2526	2565	2603	2641	2672	2703	2735	2766	2797	2827
10.0	2758	2796	2835	2873	2912	2937	2962	2987	3012	3037	3058
10.5	2967	3004	3040	3077	3114	3132	3150	3168	3186	3204	3218
11.0	3147	3177	3208	3238	3268	3280	3292	3303	3315	3327	3335
11.5	3279	3297	3314	3332	3350	3357	3363	3370	3376	3383	3387
12.0	3351	3362	3373	3384	3395	3398	3401	3405	3408	3411	3414
12.5	3393	3399	3405	3411	3417	3418	3420	3421	3423	3424	3425
13.0	3412	3415	3418	3421	3424	3425	3426	3427	3428	3429	3429
13.5	3422	3423	3425	3426	3428	3428	3429	3429	3430	3430	3430
14.0	3427	3427	3428	3428	3429	3429	3429	3430	3430	3430	3430
14.5	3428	3429	3429	3430	3430	3430	3430	3430	3430	3430	3430
15.0 to cutout ¹	3430	3430	3430	3430	3430	3430	3430	3430	3430	3430	3430

Table 2: Calculated power curve for different values of the air density for medium turbulence intensity

¹ For more details on the cut-out behaviour visit section 4 "Cut-Out and Re-Cut-In Wind Speeds"

Wind Speed at Hub Height [m/s]	Air Density										
	$\rho = 1.02$	$\rho = 1.04$	$\rho = 1.06$	$\rho = 1.08$	$\rho = 1.1$	$\rho = 1.12$	$\rho = 1.14$	$\rho = 1.16$	$\rho = 1.18$	$\rho = 1.2$	$\rho = 1.225$
	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³
3.0	32	33	35	36	38	39	40	42	43	45	46
3.5	91	94	96	99	101	104	106	109	111	114	117
4.0	163	166	170	174	178	182	186	190	194	198	203
4.5	249	255	261	266	272	278	283	289	295	300	307
5.0	354	362	369	377	385	393	400	408	416	424	433
5.5	482	492	502	513	523	533	544	554	564	574	587
6.0	634	647	660	674	687	700	713	727	740	753	770
6.5	813	830	847	864	880	897	914	931	947	964	985
7.0	1022	1043	1063	1084	1105	1126	1147	1167	1188	1209	1235
7.5	1261	1287	1312	1338	1363	1388	1414	1439	1465	1490	1521
8.0	1535	1565	1596	1626	1657	1686	1715	1745	1774	1803	1840
8.5	1840	1875	1911	1946	1981	2015	2049	2082	2116	2150	2189
9.0	2165	2203	2242	2280	2318	2354	2391	2427	2464	2500	2537
9.5	2492	2532	2571	2611	2650	2685	2719	2754	2788	2823	2854
10.0	2792	2831	2871	2910	2950	2977	3005	3032	3060	3087	3109
10.5	3016	3054	3093	3131	3170	3188	3206	3225	3243	3261	3275
11.0	3193	3225	3257	3289	3321	3331	3341	3351	3361	3371	3377
11.5	3325	3341	3358	3374	3390	3394	3399	3403	3408	3412	3414
12.0	3389	3396	3404	3411	3419	3420	3422	3423	3425	3426	3427
12.5	3417	3420	3422	3425	3428	3428	3429	3429	3430	3430	3430
13.0	3426	3427	3428	3429	3430	3430	3430	3430	3430	3430	3430
13.5	3429	3429	3430	3430	3430	3430	3430	3430	3430	3430	3430
14.0 to cutout ²	3430	3430	3430	3430	3430	3430	3430	3430	3430	3430	3430

Table 3: Calculated power curve for different values of the air density for low turbulence intensity

² For more details on the cut-out behaviour visit section 4 "Cut-Out and Re-Cut-In Wind Speeds"

Wind Speed at Hub Height [m/s]	Air Density										
	$\rho = 1.02$	$\rho = 1.04$	$\rho = 1.06$	$\rho = 1.08$	$\rho = 1.1$	$\rho = 1.12$	$\rho = 1.14$	$\rho = 1.16$	$\rho = 1.18$	$\rho = 1.2$	$\rho = 1.225$
	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³
3.0	63	65	67	69	71	73	75	77	78	80	83
3.5	126	129	132	135	139	142	145	148	151	154	158
4.0	201	205	210	215	219	224	229	233	238	243	249
4.5	291	298	304	311	317	324	330	337	343	350	358
5.0	404	412	421	430	439	447	456	465	473	482	493
5.5	541	552	563	575	586	597	609	620	632	643	657
6.0	702	717	731	746	760	775	789	804	818	833	851
6.5	891	909	927	946	964	982	999	1017	1035	1053	1074
7.0	1109	1132	1154	1177	1199	1221	1243	1264	1286	1308	1334
7.5	1355	1381	1407	1433	1459	1483	1507	1531	1555	1579	1609
8.0	1624	1654	1683	1713	1742	1770	1798	1826	1854	1882	1915
8.5	1910	1942	1975	2007	2039	2070	2100	2131	2161	2192	2225
9.0	2187	2222	2256	2291	2325	2355	2384	2414	2443	2473	2504
9.5	2461	2497	2532	2568	2603	2628	2654	2679	2705	2730	2756
10.0	2686	2720	2755	2789	2824	2845	2866	2887	2908	2929	2950
10.5	2881	2910	2940	2969	2999	3016	3033	3051	3068	3085	3101
11.0	3053	3080	3106	3133	3160	3174	3187	3201	3214	3228	3239
11.5	3186	3204	3223	3241	3260	3269	3278	3287	3296	3305	3312
12.0	3268	3281	3295	3308	3322	3329	3335	3342	3348	3355	3359
12.5	3325	3335	3346	3356	3366	3370	3375	3379	3384	3388	3391
13.0	3358	3366	3373	3381	3388	3391	3394	3398	3401	3404	3407
13.5	3384	3389	3393	3398	3402	3404	3406	3408	3410	3412	3414
14.0	3400	3403	3406	3409	3412	3413	3415	3416	3418	3419	3420
14.5	3410	3412	3415	3417	3419	3420	3421	3422	3423	3424	3424
15.0	3417	3419	3420	3422	3424	3425	3425	3426	3426	3427	3427
15.5	3418	3421	3423	3426	3428	3428	3429	3429	3430	3430	3430
16.0	3422	3424	3426	3428	3430	3430	3430	3430	3430	3430	3430
16.5	3424	3425	3427	3428	3430	3430	3430	3430	3430	3430	3430
17.0 to cutout ²	3428	3428	3429	3429	3430	3430	3430	3430	3430	3430	3430

Table 4: Calculated power curve for different values of the air density for high turbulence intensity

² For more details on the cut-out behaviour visit section 4 "Cut-Out and Re-Cut-In Wind Speeds"

3 Applicability

The power curve information provided in this document applies under the following conditions:

- The mean air density falls inside the range of air densities specified in section 2.
- The site is characterized by one of the following turbulence intensity categories.

TI Band	Lower TI bound	Upper TI bound
Low	$2.5 (0.75V_{hub} + 5.6)/V_{hub}$	$12.5 (0.75V_{hub} + 5.6)/V_{hub}$
Medium	$5 (0.75V_{hub} + 5.6)/V_{hub}$	$15 (0.75V_{hub} + 5.6)/V_{hub}$
High	$10 (0.75V_{hub} + 5.6)/V_{hub}$	$20 (0.75V_{hub} + 5.6)/V_{hub}$

Measured power curve data points for 10-minute average mean turbulence intensities that fall outside the applicable TI band would be filtered out for the purpose of power curve applicability, see Figure 1.

The reference power curve is to be selected from the power curves of the TI class having the largest number of post-filtering data points (upmost coverage). Section 6 of this document gives an example outlining how to determine the TI class with the upmost coverage. The reference power curve is referred to as the calculated power curve in GE's Technical Documentation Machine Power Performance. This document also outlines the data filtering requirements, which have to be met prior to determining the TI class.

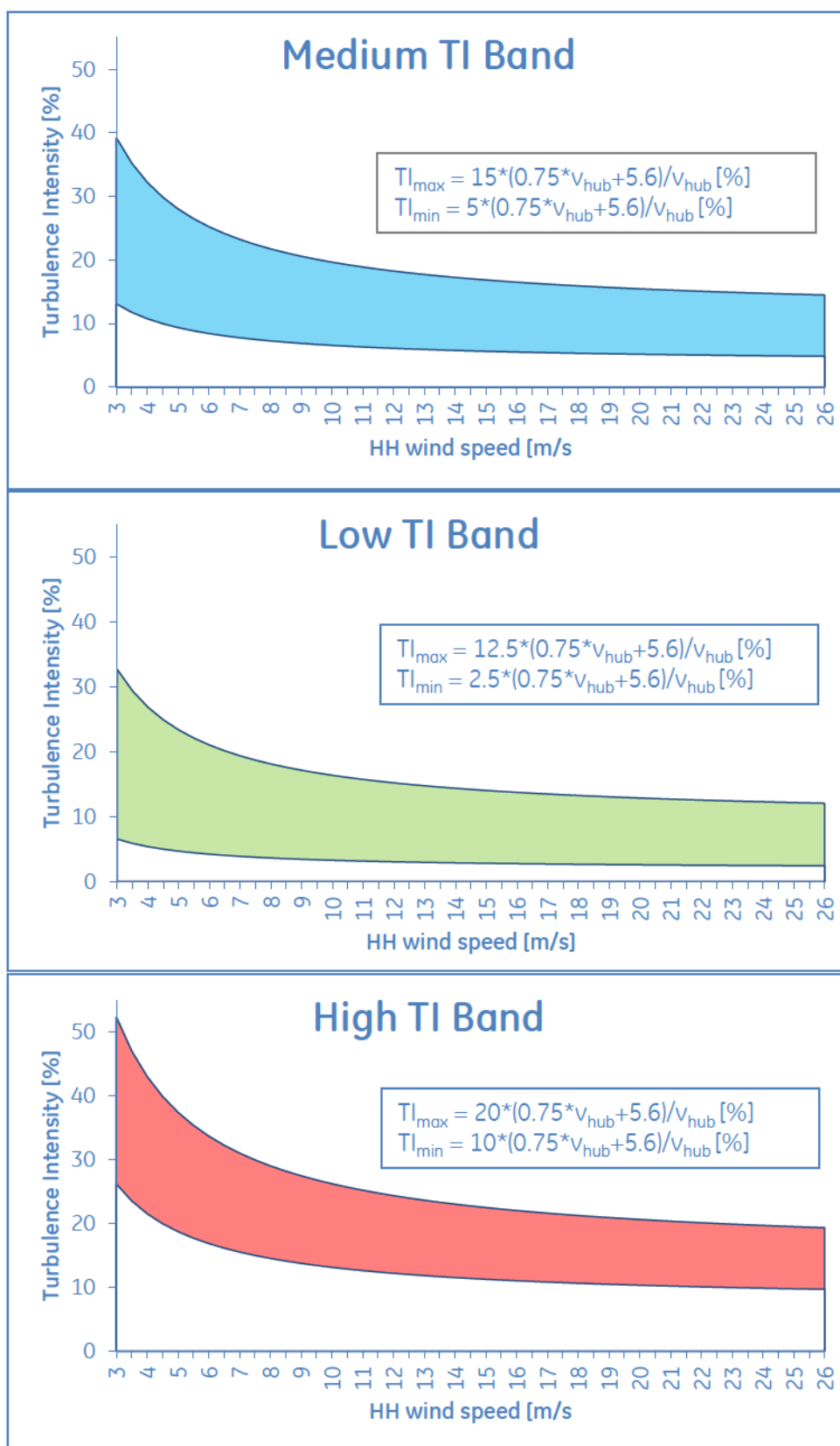


Figure 1: Applicable Turbulence Intensity Range

- Non-degraded and uncontaminated blade surfaces with no icing
- The wind turbine generator system whose power output is not being actively regulated or curtailed for any reason.
- Power shall be measured on the generator side of the transformer unless the net electric power output is referenced otherwise in this document.
- Wind-speed labels are mid-bin values; for example, the 5.0 m/s bin extends from 4.75 to 5.25 m/s.
- The flow inclination angle at the site corresponds to $-5^{\circ} < \text{flow angle} < +5^{\circ}$.
- Measured wind shear (10-minute average data) such that $-0.1 \leq \text{shear} \leq 0.5$.
- The wind turbine generator system whose power output not being curtailed due to high or low ambient temperature and/or low density. Information on the influence of cold or hot weather and high altitude operation is provided in the documents "Technical Description – Cold Weather Adaptations" and "General Description High Temperature, High Altitude Operation".

4 Cut-Out and Re-Cut-In Wind Speeds

The wind turbine generator system will shut down when the average wind speed exceeds

- 25 m/s average in a 600-second time interval.
- 30 m/s average in a 30-second time interval or
- 34 m/s average in a 3-second time interval

After the turbine has been shut-down, it will restart when the 4-minute rolling average wind speed drops to below 22m/s.

A site-specific Mechanical Loads Analysis (MLA) is required to determine the turbine suitability and whether the above-listed cut-out and re-cut-in wind speeds require adjustment to mitigate site-specific loads. If, for example, the MLA concludes that curtailment is needed to reduce loads, the MLA report will state the reduced 600-second cut-out wind speeds and the applicable wind direction sector(s). The remaining cut-out and re-cut-in wind speeds will be reduced by the same amount that the 600-second wind speed is reduced. Any reduction of the wind speeds listed in this section may be turbine location-specific. Refer to the site-specific MLA report for details.

5 Calculated Thrust Coefficient

Wind Speed at Hub Height [m/s]	Thrust Coefficient [ct]	Wind Speed at Hub Height [m/s]	Thrust Coefficient [ct]
3.0	1.00	14.5	0.19
3.5	0.90	15.0	0.17
4.0	0.87	15.5	0.15
4.5	0.85	16.0	0.14
5.0	0.84	16.5	0.13
5.5	0.83	17.0	0.12
6.0	0.83	17.5	0.11
6.5	0.83	18.0	0.10
7.0	0.83	18.5	0.09
7.5	0.82	19.0	0.09
8.0	0.79	19.5	0.08
8.5	0.75	20.0	0.08
9.0	0.70	20.5	0.07
9.5	0.65	21.0	0.07
10.0	0.58	21.5	0.06
10.5	0.52	22.0	0.06
11.0	0.45	22.5	0.06
11.5	0.39	23.0	0.05
12.0	0.34	23.5	0.05
12.5	0.30	24.0	0.05
13.0	0.26	24.5	0.05
13.5	0.23	25.0	0.04
14.0	0.21		

Table 5: Calculated thrust coefficient table; as a function of hub height wind speed

Calculated using Standard Atmospheric Conditions according to ISO 2533.

6 Guidelines for Pre-construction Energy Assessment

The following guidelines are offered for selecting power curves from this document for measurement purposes or assessment of the energy production of a new wind plant.

The air density should be the annual average air density expected at the site of the new wind plant over the life of the wind turbine.

The reference power curve for warranty purposes shall be chosen for the turbulence intensity range covering the largest fraction of data points from the measured data set of the power curve test. Figure 2 depicts one example to illustrate this guideline. The mean turbulence intensity distribution for the site is plotted over the low, medium, and high turbulence intensity ranges. As shown in Figure 2, the site has a mean turbulence intensity distribution that is most common to the low turbulence intensity range. Accordingly, the low turbulence PCs from Table 3 should be used for this example site.

For site energy assessment purposes the turbulence intensity range between low, medium, and high may also be selected based on comparing the measured 10 min averaged scatter plot in each wind speed bin expected individually to represent the conditions at the site of the new wind plant over the life of the wind turbine.

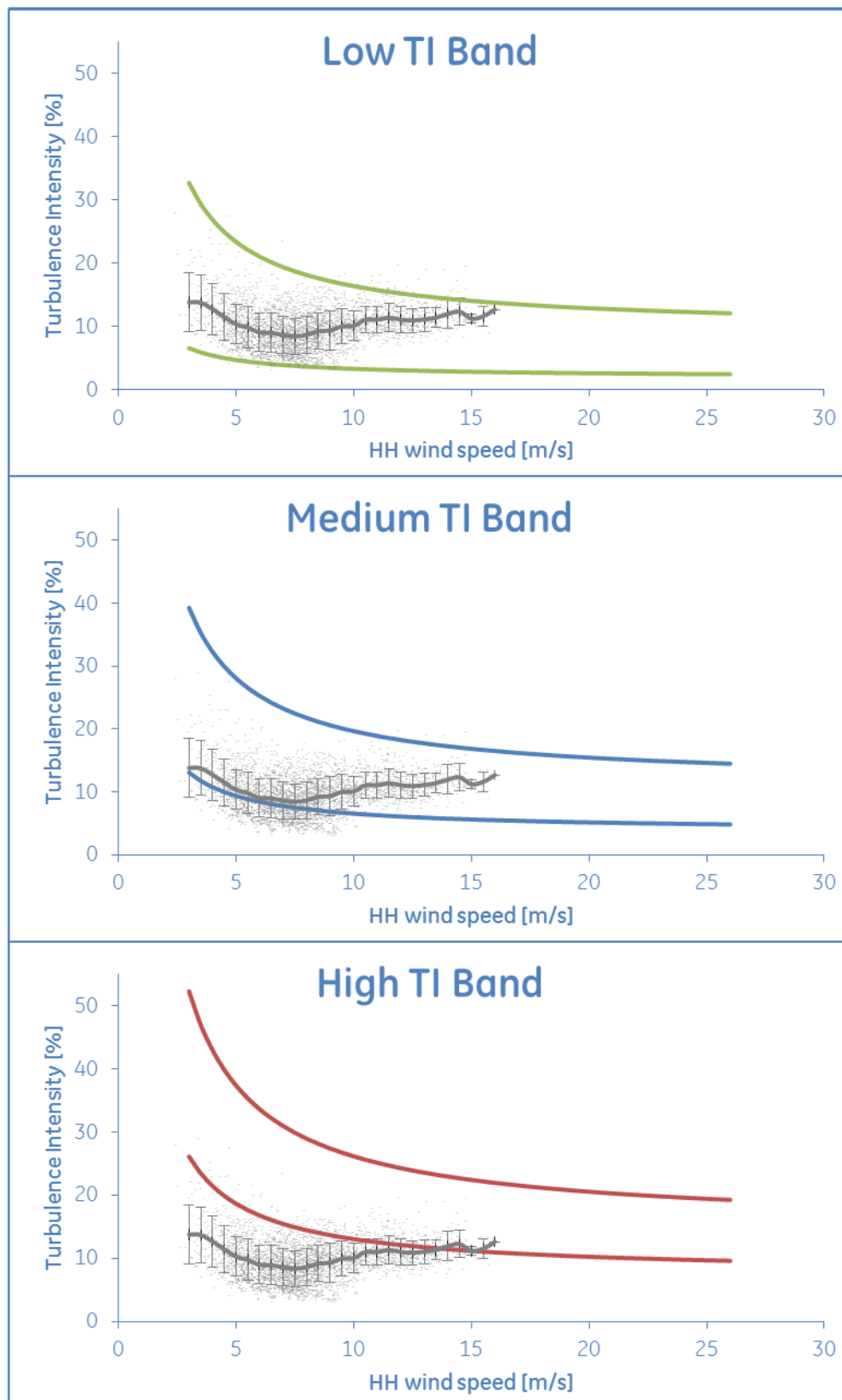


Figure 2: Example of site specific turbulence distribution with mean turbulence data and standard deviation relative to high TI (9%), medium TI (71%) and low TI (98%) band (figures in parenthesis represent the coverage rate)